

5.2.1 Definite Integral Worksheet

1. Let A be the area of the region that lies under the graph and above the x -axis of $f(x) = \ln(x)$ between $x=1$ & $x=e$. Using right endpoints, find an expression for A as a limit.

We know the following:

Width of Rectangles

$$\Delta x = \frac{e-1}{n}$$

Right Endpoints

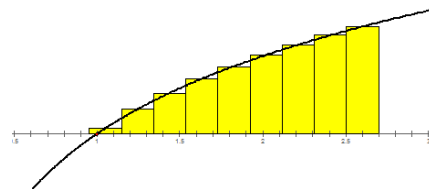
$$x_i^* = 1 + \Delta x \cdot i = 1 + \frac{e-1}{n} i$$

Height of Rectangles

$$f(x_i^*) = \ln\left(1 + \frac{e-1}{n} i\right)$$

Therefore, we can say that:

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \ln\left(1 + \frac{e-1}{n} i\right) \cdot \frac{e-1}{n}$$



2. Use the definition of the definite integral to express $\int_{-2}^2 \frac{\sin^2(x)}{x+3} dx$ as a limit.

We know the following:

Width of Rectangles

$$\Delta x = \frac{2 - (-2)}{n} = \frac{4}{n}$$

Right Endpoints

$$x_i^* = -2 + \Delta x \cdot i = -2 + \frac{4}{n} i$$

Height of Rectangles

$$f(x_i^*) = \frac{\sin^2\left(-2 + \frac{4}{n} i\right)}{-2 + \frac{4}{n} i + 3}$$

$$\int_{-2}^2 \frac{\sin^2(x)}{x+3} dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\frac{\sin^2\left(-2 + \frac{4}{n} i\right)}{1 + \frac{4}{n} i} \cdot \frac{4}{n} \right]$$

3. The area between the x -axis and the graph of $f(x) = \frac{(x+1)^2}{\ln(x)}$ on the interval $[3,15]$ can be written as a limit. Fill in the missing information in this limit so that the only variables appearing are n and i . You do not need to evaluate this limit.

We know the following:

Width of Rectangles

$$\Delta x = \frac{15-3}{n} = \frac{12}{n}$$

Right Endpoints

$$x_i = 3 + \Delta x i = 3 + \frac{12}{n} i$$

Height of Rectangles

$$f(x_i^*) = \frac{\left(3 + \frac{12}{n} i + 1\right)^2}{\ln\left(3 + \frac{12}{n} i\right)}$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[f(x_i^*) \Delta x \right] = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\frac{\left(\frac{12}{n} i + 4\right)^2}{\ln\left(3 + \frac{12}{n} i\right)} \cdot \frac{12}{n} \right]$$

4. Determine a region whose area is equal to the limit $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\sqrt{5 + \frac{2i}{n}} \cdot \frac{2}{n} \right]$. You do not need to evaluate the limit.

Width of Rectangles

$$\Delta x = \frac{2}{n} = \frac{b-a}{n}$$

Right Endpoints

$$x_i = a + \Delta x i = a + \frac{2}{n} i = a + \frac{2i}{n}$$

Function Used

$$f(x) = \sqrt{5+x}$$

We can let $b=2$ & $a=0$.

This limit represents the area under the curve $f(x) = \sqrt{5+x}$ from $[0, 2]$.

5. At time t seconds, the velocity of an object is given by the equation $v(t) = t^3$ ft/sec. The distance in feet traveled by this object from $t=12$ to $t=20$ can be written as a limit of Riemann sums in some different ways. Use your knowledge of Riemann sums to set up the following limits but do not evaluate them.

- a) Using a limit of right Riemann sums,

$$\text{Distance} = \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\left(12 + k \frac{8}{n} \right)^3 \frac{8}{n} \right]$$

- b) Using a limit of left Riemann sums,

$$\text{Distance} = \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\left(12 + (k-1) \frac{8}{n} \right)^3 \frac{8}{n} \right]$$

- c) Using a limit of midpoint Riemann sums,

$$\text{Distance} = \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\left(12 + \left(k - \frac{1}{2} \right) \frac{8}{n} \right)^3 \frac{8}{n} \right]$$

6. Evaluate the limit $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{(3k+2n)^2}{n^3}$. Use proper notation throughout.

$$\begin{aligned} \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{(3k+2n)^2}{n^3} &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{9k^2 + 12nk + 4n^2}{n^3} = \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\frac{9k^2}{n^3} + \frac{12k}{n^2} + \frac{4}{n} \right] = \lim_{n \rightarrow \infty} \left[\sum_{k=1}^n \frac{9k^2}{n^3} + \sum_{k=1}^n \frac{12k}{n^2} + \sum_{k=1}^n \frac{4}{n} \right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{9}{n^3} \sum_{k=1}^n k^2 + \frac{12}{n^2} \sum_{k=1}^n k + \frac{4}{n} \sum_{k=1}^n 1 \right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{9n(n+1)(2n+1)}{6n^3} + \frac{12n(n+1)}{2n^2} + 4 \right] \\ &= 3 + 6 + 4 \\ &= 13 \end{aligned}$$

7. Use the definition of the definite integral to write the integral $\int_2^4 (x - x^2) dx$ as a limit. Then evaluate this limit.

We know the following:

Width of Rectangles

$$\Delta x = \frac{4-2}{n} = \frac{2}{n}$$

Right Endpoints

$$x_i = 2 + \Delta x i = 2 + \frac{2}{n} i$$

Height of Rectangles

$$f(x_i^*) = \left(2 + \frac{2}{n} i\right) - \left(2 + \frac{2}{n} i\right)^2$$

$$\int_2^4 (x - x^2) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\left[\left(2 + \frac{2}{n} i\right) - \left(2 + \frac{2}{n} i\right)^2 \right] \frac{2}{n} \right]$$

So we work to evaluate

$$\begin{aligned} & \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\left[\left(2 + \frac{2}{n} i\right) - \left(2 + \frac{2}{n} i\right)^2 \right] \frac{2}{n} \right] \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\left[\left(2 + \frac{2}{n} i\right) - \left(4 + \frac{8}{n} i + \frac{4}{n^2} i^2\right) \right] \frac{2}{n} \right] = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\left(-2 - \frac{6}{n} i - \frac{4}{n^2} i^2 \right) \frac{2}{n} \right] = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[-\frac{4}{n} - \frac{12}{n^2} i - \frac{8}{n^3} i^2 \right] \\ &= \lim_{n \rightarrow \infty} \left[\sum_{i=1}^n \left(-\frac{4}{n} \right) - \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{12}{n^2} i \right) - \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{8}{n^3} i^2 \right) \right] \quad \text{by properties 6 \& 7} \\ &= \lim_{n \rightarrow \infty} \left[\left(-\frac{4}{n} \right) n - \frac{12}{n^2} \sum_{i=1}^n i - \frac{8}{n^3} \sum_{i=1}^n i^2 \right] \quad \text{by properties 4 \& 5} \\ &= \lim_{n \rightarrow \infty} \left[-4 - \frac{12}{n^2} \cdot \frac{n(n+1)}{2} - \frac{8}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} \right] \quad \text{by properties 1 \& 2} \\ &= \lim_{n \rightarrow \infty} \left[-4 - \frac{6n^2 + 6n}{n^2} - \frac{8n(n+1)(2n+1)}{6n^3} \right] = \lim_{n \rightarrow \infty} \left[-4 - \frac{6n^2 + 6n}{n^2} - \frac{16n^3 + 24n^2 + 8n}{6n^3} \right] \\ &= -4 - 6 - \frac{16}{6} \\ &= -\frac{38}{3} \end{aligned}$$

Therefore, we have found that:

$$\int_2^4 (x - x^2) dx = -\frac{38}{3}$$

8. Use the definition of the integral to evaluate $\int_1^5 (x^2 + 3x + 2) dx$.

We know the following:

Width of Rectangles

$$\Delta x = \frac{5-1}{n} = \frac{4}{n}$$

Right Endpoints

$$x_i = 1 + \Delta x i = 1 + \frac{4}{n} i$$

Height of Rectangles

$$f(x_i^*) = \left(1 + \frac{4}{n} i\right)^2 + 3\left(1 + \frac{4}{n} i\right) + 2$$

$$\int_1^5 (x^2 + 3x + 2) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\left[\left(1 + \frac{4}{n} i\right)^2 + 3\left(1 + \frac{4}{n} i\right) + 2 \right] \frac{4}{n} \right]$$

So we work to evaluate

$$\begin{aligned} & \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\left[\left(1 + \frac{4}{n} i\right)^2 + 3\left(1 + \frac{4}{n} i\right) + 2 \right] \frac{4}{n} \right] \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\left[1 + \frac{8}{n} i + \frac{16}{n^2} i^2 + 3 + \frac{12}{n} i + 2 \right] \frac{4}{n} \right] = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\left(6 + \frac{20}{n} i + \frac{16}{n^2} i^2 \right) \frac{4}{n} \right] = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\frac{24}{n} + \frac{80}{n^2} i + \frac{64}{n^3} i^2 \right] \\ &= \lim_{n \rightarrow \infty} \left[\sum_{i=1}^n \left(\frac{24}{n} \right) + \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{80}{n^2} i \right) + \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{64}{n^3} i^2 \right) \right] \quad \text{by properties 6 \& 7} \\ &= \lim_{n \rightarrow \infty} \left[\left(\frac{24}{n} \right) n + \frac{80}{n^2} \sum_{i=1}^n i + \frac{64}{n^3} \sum_{i=1}^n i^2 \right] \quad \text{by properties 4 \& 5} \\ &= \lim_{n \rightarrow \infty} \left[24 + \frac{80}{n^2} \cdot \frac{n(n+1)}{2} + \frac{64}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} \right] \quad \text{by properties 1 \& 2} \\ &= \lim_{n \rightarrow \infty} \left[24 + \frac{40n^2 + 40n}{n^2} + \frac{64(n+1)(2n+1)}{6n^3} \right] = \lim_{n \rightarrow \infty} \left[24 + \frac{40n^2 + 40n}{n^2} + \frac{128n^3 + 192n^2 + 64n}{6n^3} \right] \\ &= 24 + 40 + \frac{128}{6} \\ &= \frac{256}{3} \end{aligned}$$

Therefore, we have found that:

$$\int_1^5 (x^2 + 3x + 2) dx = \frac{256}{3}$$